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Matching in Dynamic Imbalanced Markets

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1 Big picture

- **Question.** In a dynamic matching market with easy-to-match and hard-to-match agents, should a platform match greedily as soon as compatible agents appear, or wait to thicken the market?
- **Motivation.** Kidney exchange platforms face a trade-off between faster transplants and thicker matching pools.
- **Core insight.** In large imbalanced markets, the apparent trade-off vanishes. Greedy matching both minimizes waiting times and maximizes match rates asymptotically.
- **Main theorem.** Greedy is asymptotically optimal; fixed batching and patient policies are not.
- **Empirical confirmation.** Simulations using National Kidney Registry data show that greedy matching achieves similar or better match rates and shorter waiting times than patient and batching policies.

2 Incremental contribution of the paper

- The paper identifies a setting in which the classic thickness-versus-speed trade-off disappears.
- It introduces a two-type random compatibility model that matches the empirical kidney-exchange facts that SMM stays below one while FWP approaches zero.
- It proves that homogeneous random graph models cannot match both facts.
- It derives closed-form large-market match rates and waiting times for greedy, batching, and patient policies.
- It validates the theory using real National Kidney Registry compatibility data.

3 Model objects and measurement

There are easy-to-match agents E and hard-to-match agents H . If there are m easy agents, there are $(1+\lambda)m$ hard agents. Easy-hard pairs are compatible with probability $p > 0$, easy-easy pairs with probability $q > 0$, and hard-hard pairs are incompatible.

4 Models and theoretical specifications

4.1 Static model

$$SMM \rightarrow \frac{2}{2+\lambda}, \quad FWP \rightarrow 0.$$

Detailed interpretation: The model captures the fact that almost everyone has some compatible partner, but not everyone can be matched simultaneously because easy agents are scarce.

4.2 Greedy policy

$$q_H^G = \frac{1}{1+\lambda}, \quad q_E^G = 1, \quad w_H^G = \frac{\lambda d}{1+\lambda}, \quad w_E^G = 0.$$

Detailed interpretation: Easy agents match immediately to waiting hard agents. Hard agents are limited by the scarce supply of easy agents.

4.3 Batching policy

$$q_H^B = \frac{1 - e^{-T/d}}{(1 + \lambda)T/d}, \quad q_E^B = \frac{1 - e^{-T/d}}{T/d}, \quad w_\theta^B = d(1 - q_\theta^B).$$

Detailed interpretation: Fixed batching imposes delay and causes some agents to become critical before matching.

4.4 Patient policy

$$q_H^P = \frac{1}{1 + \lambda}, \quad q_E^P = 1, \quad w_H^P = d, \quad w_E^P = 0.$$

Detailed interpretation: Patient matching preserves match rates but makes hard agents wait too long.

5 Identification, theoretical design, and mechanism credibility

The model is disciplined by static NKR compatibility facts. The core mechanism is that hard-to-match agents accumulate in large imbalanced markets, so greedy does not destroy the relevant liquidity. NKR simulations confirm the main predictions in richer finite markets with blood types, antigens, and antibodies.

6 Section-by-section study guide

- **Introduction.** Explains the practitioner concern about greedy matching and states the main result.
- **Static model.** Shows why two types are needed.
- **Dynamic model.** Defines arrivals, departures, policies, match rates, and waiting times.
- **Large-market theory.** Proves greedy optimality and derives limits for batching and patient policies.
- **Finite-market analysis.** Quantifies how frequent batching must be.
- **Empirical section.** Uses NKR data to validate predictions.

7 Tables and figures

7.1 Table 1: Fraction of pairs matched and average waiting times in days over all pairs in simulations using NKR data

Read: Greedy has the shortest waiting time at every arrival rate. Patient has slightly higher match rates but much longer waiting times. Longer batching intervals reduce match rates and raise waiting times.

Detailed interpretation: Table 1 is the main empirical policy comparison. It shows why match rates alone are insufficient: patient matching can produce a small match-rate gain but at large waiting-time cost.

Economic message: Greedy achieves the best overall policy performance when both transplants and waiting costs matter.

TABLE 1
Fraction of pairs matched and average waiting times in days over all pairs in simulations using NKR data

Arrivals per day	Match rate (%)					Waiting time in days				
	Greedy	Patient	Batching 7 days	30 days	60 days	Greedy	Patient	Batching 7 days	30 days	60 days
0.01	10.7	11.9	10.4	9.9	9.3	322	355	322	324	326
0.05	22.4	23.4	22.2	21.2	20.2	279	324	280	283	288
0.25	34.3	35.4	33.8	32.6	31.2	237	298	238	243	248
0.5	38.5	39.5	38.0	36.8	35.2	222	290	223	228	233
1	42.0	43	41.6	40.2	38.6	209	283	210	215	221
2	45	45.8	44.5	43.1	41.5	198	278	200	205	211
4	47.2	48	46.8	45.4	43.6	190	274	192	196	207

Our theory combined with simulations suggest that matching greedily is not a real source of inefficiency.

Table 1: NKR simulation match rates and waiting times

7.2 Figure 1 and Figure 2: Empirical graph facts and random compatibility model

Read: Figure 1 shows that SMM stays below one while FWP falls toward zero. Figure 2 shows the two-type compatibility graph.

Detailed interpretation: The figures justify the model. Heterogeneity and imbalance are needed to explain the static compatibility facts.

Economic message: The market is not simply sparse; it is imbalanced.

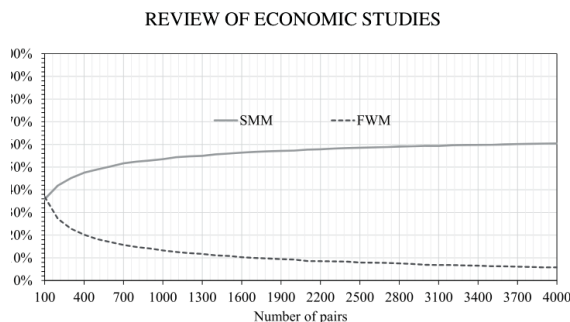


FIGURE 1

Figure 1: The percentage of pairs without a compatible partner (dashed) and the percentage matched in a maximum matching (solid) as a function of the number of pairs. The average for every fixed pool size on the horizontal axis is computed by random sampling from the population.

(a) Figure 1: FWP and SMM

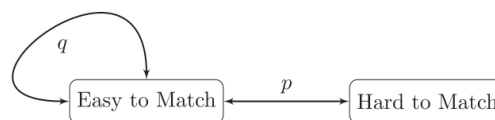


FIGURE 2

The random compatibility model.

Figure 2: The random compatibility model. In this model, each patient and donor are virtually compatible, and therefore the matching problem is solved by a greedy algorithm. Two features stand out: first, even as the market grows large, the SMM remains below one, and second, the FWP remains above zero.

(b) Figure 2: random compatibility model

7.3 Figure 3: Illustration of Proposition 3

Read: Greedy lies near the high-match, low-waiting frontier. Batching is worse on both dimensions; patient has similar match rates but much longer waiting for hard agents.

Detailed interpretation: Figure 3 is the visual proof of the large-market comparison.

Economic message: The apparent thickness-speed trade-off vanishes in large imbalanced markets.

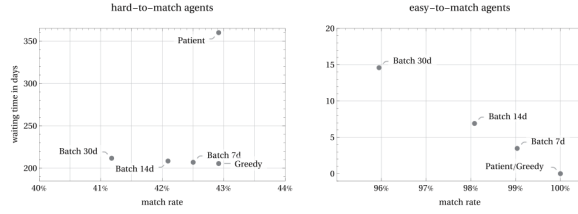


FIGURE 3
Illustration of Proposition 3 when $\lambda = 1.33$ and d equals 360 days. The points represent the predictions of our model for large markets, which are derived in Proposition 3.

Figure 3: Proposition 3 match rates and waiting times

7.4 Figure 4 and Figure 5: Finite-market comparison of greedy and batching

Read: Figure 4 gives a batch-length bound above which greedy dominates. Figure 5 shows simulations across market sizes, imbalance values, and compatibilities.

Detailed interpretation: Batching must be very frequent to avoid being dominated by greedy in realistic kidney-exchange-like markets.

Economic message: Waiting for monthly batch runs is too slow when agents can become critical.

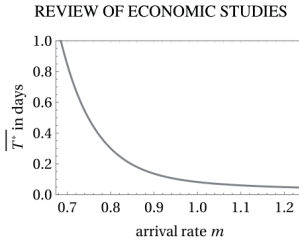
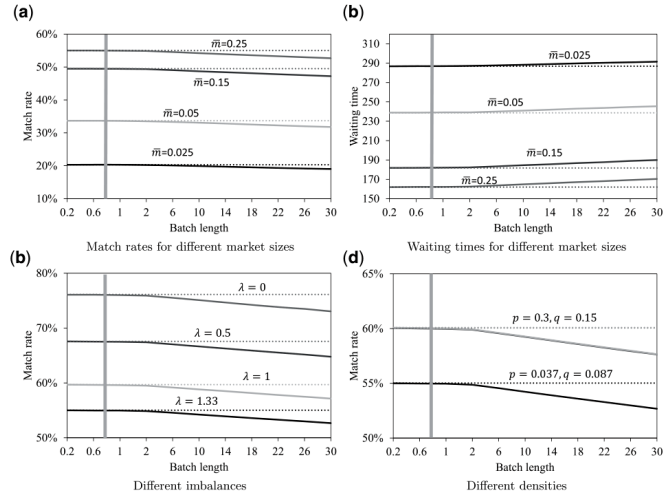


FIGURE 4
above which greedy dominates batching for various arrival rates per day, $\lambda = 1.33, p$

(a) Figure 4: batching bound



(b) Figure 5: greedy versus batching

7.5 Figure 6: Extreme match probabilities

Read: In an extreme finite example, batching can outperform greedy when the market is balanced and easy-easy compatibilities are high.

Detailed interpretation: The figure shows the limits of the theorem and clarifies that greedy is not universally dominant for every finite parameterization.

Economic message: The calibrated kidney-exchange environment is not the special case where batching is useful.

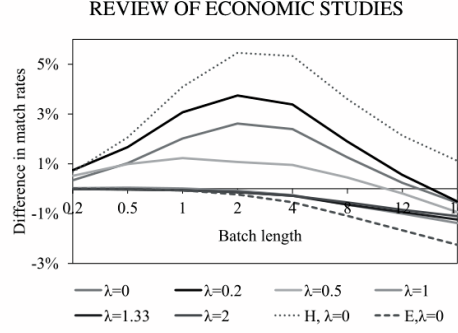


FIGURE 6

Figure 6: extreme compatibility counterexample

7.6 Figures 7–9: NKR simulations

Read: Figure 7 compares greedy and batching in NKR data. Figure 8 compares greedy and patient policies. Figure 9 compares waiting times and match rates pair by pair.

Detailed interpretation: The NKR data confirm the theory: greedy has similar match rates and shorter waiting times. The pair-level comparison shows greedy does not systematically harm hard-to-match categories.

Economic message: Greedy matching is practically attractive in the real kidney exchange environment.

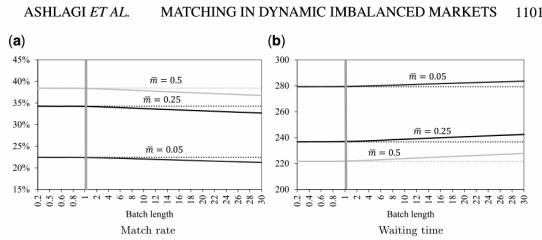


FIGURE 7

(a) Figure 7: greedy versus batching

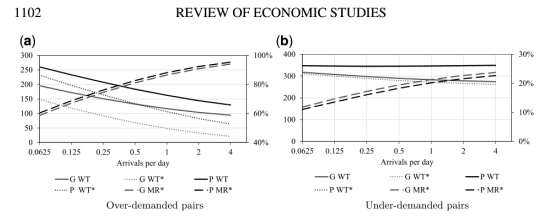


FIGURE 8

Average waiting times (WT) and match rate (MR) in days under greedy (G) and patient (P) policies. The left and right

(b) Figure 8: greedy versus patient

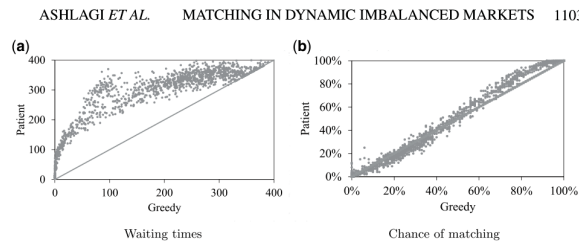


FIGURE 9

Averages of waitine times (left) and chance of matchine (right) taken over conies for each pair in the data. The axes

Figure 9: pair-level greedy versus patient comparison

8 Short summary

- The paper studies dynamic matching in imbalanced exchange markets.
- A two-type model captures kidney-exchange facts that homogeneous models cannot.
- Greedy is asymptotically optimal in large imbalanced markets.
- Fixed batching loses on match rates and waiting times.
- Patient matching preserves match rates but makes hard agents wait longer.
- NKR simulations support the theory.

9 Expanded conclusion

9.1 Core conclusion

Greedy matching is not merely a heuristic in large imbalanced markets. It is asymptotically optimal and empirically attractive in kidney-exchange simulations.

9.2 Economic intuition

Imbalance creates endogenous thickness. Hard agents accumulate in the pool, so arriving easy agents can be matched immediately with hard agents.

9.3 Incremental contribution

The paper identifies conditions under which waiting to thicken the market is unnecessary and potentially harmful. It combines graph theory, dynamic matching theory, and NKR simulations.

9.4 Policy implication

Kidney exchange platforms should not assume that frequent matching is harmful. Greedy matching can reduce waiting times without lowering transplant rates.

9.5 Limitations

The theory abstracts from altruistic chains, strategic reporting, and full biological heterogeneity, though NKR simulations partially address heterogeneity.

9.6 Takeaway

In large imbalanced markets, faster matching can be better matching.